

Raising the apex raises the Gaussian centroid

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Abstract

Restrict the standard Gaussian measure to a triangle and take its centre of mass. Does the centre of mass determine the triangle when only one vertex may move? We prove the case in which the moving vertex rises along the perpendicular to the opposite side through a point of that side: the Gaussian centroid then rises strictly, in every dimension and over any convex base, wherever the Gaussian is centred. The proof is a monotone likelihood ratio, supplied by Prékopa's theorem on log-concave functions. The general case, in which the vertex moves obliquely and one triangle contains the other, remains open. The steps of the proof after Prékopa's theorem are checked in the Lean proof assistant.

1 The question

Let γ be the standard Gaussian density on $\mathbb{R}^2$. For a triangle T its Gaussian centroid is

$$e(T) = \frac{\int_T x \gamma(x) dx}{\int_T \gamma(x) dx},$$

the centre of mass of the Gaussian restricted to T . On MathOverflow [1], jens asked whether two triangles ABC and ABC' with $C' \neq C$ always have different Gaussian centroids, and noted that he needs the same for n -simplices in $\mathbb{R}^n$. If neither triangle contains the other, a line separates what is removed from what is added and the centroid shifts across it, as he observes. The hard case is when one triangle contains the other: the bigger one has extra mass, but the extra mass is far out, where the Gaussian is small, and it is not obvious that the centroid must move.

Here is the simplest nested case to try before reading on. Keep A and B fixed, put the apex at height h above a point D of the segment AB , and raise it. The triangle grows upwards. Does its Gaussian centroid rise? For the uniform density the centroid height is $h/3$, but the Gaussian may be centred anywhere, even far to one side, and the new mass lies where the density is small.

Theorem 1. *Let K be a compact convex set with nonempty interior in a hyperplane H of $\mathbb{R}^n$, $n \geq 2$, let $D \in K$, and let ν be a unit normal to H . For $h > 0$ let T_h be the convex hull of K and the apex $D + h\nu$. Under the standard Gaussian restricted to T_h , the mean distance from H is strictly increasing in h .*

Corollary 1. *If C and C' are distinct points on the line through a point D of the segment AB perpendicular to AB , and neither equals D , then ABC and ABC' have different Gaussian centroids. The same holds for simplices, with the apex moving along the normal to the opposite facet through a point of that facet.*

Proof. On one side of AB apply Theorem 1 with K the segment AB . If C and C' are on opposite sides, the two centroids are on opposite sides of the line AB , since each triangle lies on its own side. $\square$

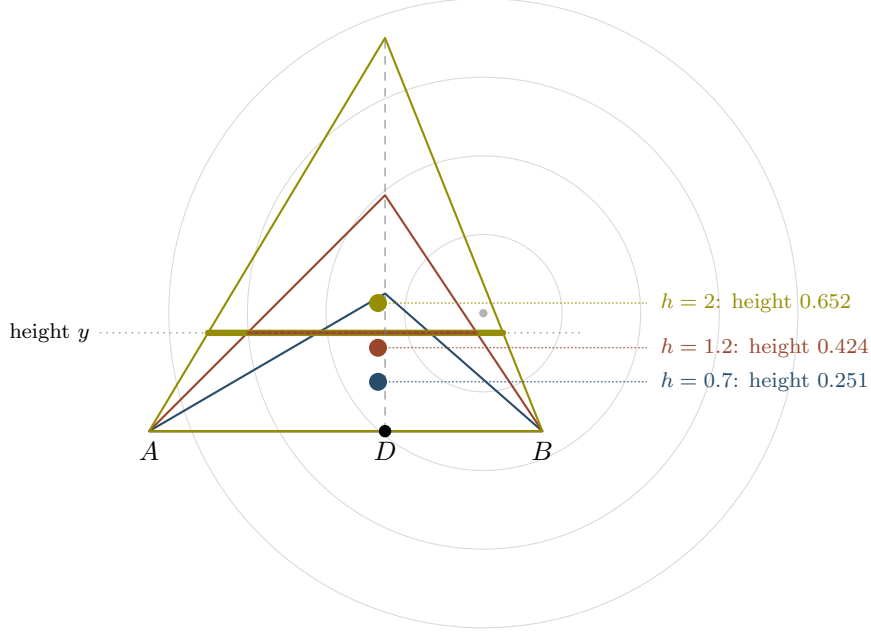


Figure 1: Three apexes above the foot D and the Gaussian contours, centred off the axis of symmetry. The slice at height y of each triangle is the base shrunk towards D by the factor $1 - y/h$. The Gaussian centroids (dots, computed by quadrature) rise with h .

The theorem needs no symmetry of the base or of the Gaussian: D may be anywhere in K , and the Gaussian may be centred anywhere. Two of its hypotheses cannot be dropped. If the foot is outside the base, the mean height can fall: for $A = (10, 0)$, $B = (11, 0)$ and apex $(0, h)$, the mean height is about 5.050 at $h = 10$ and 4.048 at $h = 20$. And the proof uses log-concavity in an essential way, as the remark after Lemma 2 explains.

2 Proof

Take coordinates (x, y) with $x \in H$ and y the signed distance from H . The standard Gaussian density factors as $\rho(x)w(y)$, with ρ a Gaussian density on H and w a one-dimensional one, because H and ν are orthogonal. The section of T_h at height $y \in (0, h)$ is the base shrunk towards D :

$$D + s(K - D), \quad s = 1 - \frac{y}{h}.$$

So the height of a Gaussian point of T_h has density proportional to

$$q_h(y) = w(y) F\left(1 - \frac{y}{h}\right) \quad (0 < y < h), \quad F(s) = \int_{D+s(K-D)} \rho(x) dx.$$

Read $F(s)$ as the Gaussian mass of the base scaled by s about D . Everything depends on two properties of F .

F is strictly increasing. As $D \in K$ and K is convex, the scaled copies are nested, and their volumes $s^{n-1} \text{vol}(K)$ grow strictly; ρ is positive.

$\log F$ is concave.

Lemma 1. For $s_1, s_2 > 0$, $0 \leq \lambda \leq 1$ and $z_1, z_2 \in K$,

$$(1 - \lambda)(D + s_1(z_1 - D)) + \lambda(D + s_2(z_2 - D)) = D + s(z - D)$$

for $s = (1 - \lambda)s_1 + \lambda s_2$ and some $z \in K$.

Proof. Take $z = az_1 + bz_2$ with $a = (1 - \lambda)s_1/s$ and $b = \lambda s_2/s$, which are nonnegative and sum to 1; then $s(z - D) = (1 - \lambda)s_1(z_1 - D) + \lambda s_2(z_2 - D)$. $\square$

So the set $\mathcal{C} = \{(x, s) : s > 0, x \in D + s(K - D)\}$ is convex, the function $(x, s) \mapsto \rho(x)\mathbf{1}_{\mathcal{C}}(x, s)$ is log-concave, and by Prékopa's theorem [2, Theorem 6] its integral over x , which is $F(s)$, is log-concave.

Lemma 2. *Let φ be concave and strictly increasing on $(0, \infty)$, and $0 < h_1 < h_2$. Then*

$$y \mapsto \varphi\left(1 - \frac{y}{h_2}\right) - \varphi\left(1 - \frac{y}{h_1}\right)$$

is strictly increasing on $(0, h_1)$.

Proof. Let $y < y'$. As y moves to y' , the argument $1 - y/h_1$ runs back over an interval of length $(y' - y)/h_1$, and $1 - y/h_2$ over a shorter interval, of length $(y' - y)/h_2$, that starts no lower. A concave function gains no more over an interval of given length the further right it lies: writing u_2 and $u_1 + L$ as complementary convex combinations of u_1 and $u_2 + L$ and adding the two concavity inequalities gives $\varphi(u_2 + L) - \varphi(u_2) \leq \varphi(u_1 + L) - \varphi(u_1)$ for $u_1 \leq u_2$. A strictly increasing function gains strictly more over a longer interval with the same left end. Together, φ loses strictly more along the first argument than along the second, which is the claim. $\square$

Apply it with $\varphi = \log F$: the likelihood ratio $r(y) = q_{h_2}(y)/q_{h_1}(y) = F(1 - y/h_2)/F(1 - y/h_1)$ is strictly increasing on $(0, h_1)$. The factor $w(y)$ has cancelled, which is why the Gaussian may be centred anywhere.

Remark. Without log-concavity the ratio need not increase. For $F(s) = se^{50s^2}$, which is increasing, the derivative of $\log r$ at $h_1 = 1$, $h_2 = 2$, $y = \frac{4}{5}$ is $-\frac{35}{6}$.

Proof of Theorem 1. Let $0 < h_1 < h_2$ and let μ be the height law for h_1 , with density proportional to q_{h_1} on $(0, h_1)$. The law for h_2 , conditioned on $(0, h_1)$, has density proportional to $r q_{h_1}$, so its mean is $\int y r d\mu / \int r d\mu$. Chebyshev's integral inequality for the two increasing functions y and r says

$$\int y d\mu \int r d\mu < \int y r d\mu,$$

strictly because $(y - y')(r(y) - r(y')) > 0$ whenever $y \neq y'$, and $\mu \otimes \mu$ gives the diagonal no mass. (The proof is one line: the double integral of $(y - y')(r(y) - r(y'))$ is positive and equals $2(\int y r - \int y \int r)$ for a probability measure.) So the conditional mean exceeds the mean for h_1 . The rest of the h_2 law lies in $[h_1, h_2)$, above both, so the mean for h_2 is larger still. $\square$

Remark. The proof gives more than the means: the height laws increase in the likelihood-ratio order. It also works for any positive log-concave density in place of ρ and any positive weight $w(y)$, which cancels. It does not reach the general nested case. When the apex moves obliquely, the sections are no longer scaled copies of the base about a fixed point, so the height laws are not all built from one function F , and the likelihood-ratio comparison has nothing to hold on to. Whether the Gaussian centroid determines the moving vertex in general remains open.

Verification. Lemma 1, the increment inequality and Lemma 2, Chebyshev's integral inequality with its strict form for any finite measure on $\mathbb{R}$, and the final comparison of means are checked in Lean 4 with Mathlib [3], using only the standard axioms. Quoted rather than formalised: Prékopa's theorem, the factorisation of the Gaussian and the resulting height densities, and the strict increase of F . A Python program audits the section identities, the likelihood ratios, the means and the two counterexamples by quadrature, and deliberately wrong variants

fail it; an independent program, sharing no code with it, computes Gaussian centroids of triangles and of a tetrahedron by Monte Carlo and checks that they rise with the apex. Everything is at <https://github.com/jvvk/mathematics/tree/main/raising-the-apex>.

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References

- [1] jens, Gaussian expectation on a triangle is one-to-one with any single vertex?, MathOverflow question 499635 (26 August 2025), <https://mathoverflow.net/q/499635>, accessed 10 October 2026.
- [2] A. Prékopa, On logarithmic concave measures and functions, *Acta Sci. Math. (Szeged)* **34** (1973) 335–343.
- [3] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381, doi:10.1145/3372885.3373824.

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